Active Learning for AMR Model Retraining — Data, Gap and Publication Plan

Dyen Asif · 24I-0608 · FAST-NUCES Islamabad · prepared for meeting of 24 Sep 2026 · figures marked ≈ must be verified before citing

1. The data question, answered

One data type throughout: MALDI-TOF mass spectra + lab-confirmed S/I/R labels, from DRIAMS. Training and testing use the same kind of data. What changes between them is **time** (and, in the extension, **hospital**) — never the data type. That shift is the thing the project studies.

Role	Data	Label status
Initial training set	DRIAMS-A (University Hospital Basel), 2015–2016	labelled
Arrival pool (what active learning chooses from)	DRIAMS-A, 2017, released one quarter at a time	labels hidden; revealed only for isolates the method "sends to the lab"
Test set	DRIAMS-A, 2018	never used for training or selection
External test (extension)	DRIAMS-B, C, D — other Swiss labs, 2018	held out; measures cross-site shift, same data type

- **Input features:** each spectrum binned to a fixed-length vector (standard DRIAMS preprocessing, 2,000–20,000 Da). **Output:** resistant vs susceptible for one species–antibiotic pair.
- **Pairs:** the three used in the DRIAMS paper — *E. coli*–ceftriaxone, *K. pneumoniae*–ceftriaxone, *S. aureus*–oxacillin. One model per pair.
- **How "lab cost" is simulated:** every DRIAMS isolate was actually tested, so all labels exist. Active learning is run as standard pool-based simulation: a label counts as "paid for" only when the method asks for it. Cost = number of labels revealed.
- **Scale:** DRIAMS has >300,000 spectra and >750,000 resistance labels across four sites; DRIAMS-A alone is roughly 38,000 spectra in the published splits. Plenty for quarterly pools.

> **Two corrections to the slides.**
>
> (1) Slide 15 pairs DRIAMS (spectra) with BV-BRC (genomes) without saying how. A spectrum model cannot be tested on genomes. Recommendation: **drop BV-BRC from the core study.** If kept, it is a separate, self-contained second experiment with its own train/test split, reported separately.
>
> (2) Slide 15 says DRIAMS spans "four hospitals over several years". Only DRIAMS-A spans multiple years (2015–2018); B–D cover 2018. The time-based split must therefore use DRIAMS-A.

2. Research gap — assessment

The gap holds. Decay of MALDI-TOF AMR models is measured (Wiesmann 2025: AUROC −0.10 to −0.25 in 18 months) and its authors prescribe regular retraining without costing it. Active learning with a real wet-lab oracle is proven (Wu 2026). A search on 24 Sep found no work applying active learning to choose which clinical isolates get phenotyped for retraining. That is a quick search, not a systematic review — do a proper one (PubMed, Scopus, Google Scholar, bioRxiv) before writing "never been done" in a paper.

Three improvements worth making — each fixes a real weakness, none is decoration:

1. **Make RQ2 match the data.** RQ2 and slide 14 talk about "genetically clustered" isolates and "lineages", but DRIAMS has no genomes. In the spectra setting, diversity means *spectral* diversity — clusters of similar spectra,

which partly track strain type. Reword RQ2 to: "which selection strategy works best when isolates arrive in clusters of near-identical spectra?" Same idea, now testable on the data.

2. **Check the premise first (month 2).** Decay was measured in a German hospital, not in DRIAMS. If DRIAMS-A shows little drop from 2015–16 to 2018, the retraining story weakens. Fallback already exists in the same dataset: **adapting to a new hospital** (train on A, arrive at B/C/D with a small labelling budget) — cross-site drops are large and documented (Duroux 2026: AUPRC down up to 0.44).
3. **Promote the cross-site setting from "future work" to RQ4.** "A hospital adopts a model trained elsewhere: how few local isolates must it test to recover performance?" This is clinically concrete, uses the same data type, and is the strongest route to a ≥5 IF paper because it gives two shift scenarios (time and site) instead of one.

Evaluation (unchanged in spirit, made explicit): AUROC and AUPRC on the 2018 test set; very major error rate (reporting S when truly R — the dangerous error) and major error rate; accuracy-versus-labels curves for random, uncertainty, diversity and hybrid selection at matched budgets; 5+ random seeds with confidence intervals. Headline number: **labels needed to recover 95% of full-retraining AUROC**, per strategy.

3. Target journals (IF ≥ 5)

Impact factors are approximate from memory and change yearly — confirm each on Clarivate JCR (via the FAST library) before the meeting slide or any application.

Journal	IF ≈	Fit	Cost to publish
Computers in Biology and Medicine (Elsevier)	7	Strong — applied ML on biomedical data	Free via subscription route; OA APC ≈ USD 3,080
IEEE J. Biomedical and Health Informatics	6–7	Strong — ML methods for clinical data	Free via standard route (≤8 pages); OA APC ≈ USD 2,800
Briefings in Bioinformatics (OUP)	7–9	Good if framed as a methods/benchmark study	**Fully open access since 2024 — APC ≈ USD 3,885** (LMIC waiver/discount possible)
Artificial Intelligence in Medicine (Elsevier)	6–7	Good	Free via subscription route
Journal of Clinical Microbiology (ASM)	5–6	Good for the clinical audience (Wiesmann published here)	Check ASM author fees
Nature Communications / Clin. Microbiol. Infection	14 / 9	Stretch — needs real-lab or multi-site validation	Nat Comms is fully OA: APC in the thousands of USD

Realistic first target: Computers in Biology and Medicine or IEEE JBHI. **Also check slide 11:** the IF of 13.6 for *npj Antimicrobials and Resistance* looks wrong — it is a recently launched journal and may not have an impact factor yet.

> **"Is every journal funded?" — no, it works the other way.** Journals do not pay authors. **Subscription/hybrid journals** (all in the table except Briefings in Bioinformatics and Nature Communications) are **free to publish in** if you do not choose open access. **Fully open-access journals** charge the author an article processing charge (APC). Many publishers give APC waivers or discounts to authors from lower-middle-income countries such as Pakistan — check each journal's waiver page. For a first paper, the subscription route costs nothing.

4. Funding options at FAST

Source	What it covers	Who applies
FAST ORIC (Office of Research, Innovation & Commercialization)	Internal research grants, publication/APC support if offered	Ask ORIC — first stop
HEC NRPU (National Research Program for Universities)	Project grants: equipment, RA stipends, compute	Faculty member as PI (a supervisor), student as RA
HEC travel grant	Conference travel to present the work	Faculty / student, per HEC rules
Pakistan Science Foundation	Research project grants	Faculty PI
Cloud research credits (Google, AWS, Microsoft)	Compute, if needed	Student or faculty

Honest scale: this project needs very little money. DRIAMS is free, the models are small (a laptop or free Colab is enough), and the paper can go to a subscription journal at no cost. Funding matters mainly if you want open access, conference travel, or real-lab validation with a local hospital.

5. Questions to ask in the meeting

1. Is DRIAMS, MALDI-TOF only, acceptable as the single dataset — or do you want the genome (BV-BRC) route as well, as a separate experiment?
2. Do you agree with the split: train 2015–16, arrival pool 2017, test 2018, with DRIAMS-B/C/D as external test?
3. Should cross-site adaptation be a main research question (RQ4) or stay as future work?
4. Which species–antibiotic pairs matter most to you — the three standard ones, or any with local relevance in Pakistan?
5. Is there any chance of a local hospital lab (e.g. in Islamabad/Rawalpindi) sharing MALDI-TOF data later for validation? That would lift the paper a tier.
6. Which journal do you want to aim for first, and who will be corresponding author?
7. Would either of you apply for NRPU or an ORIC grant with this as the project? Does FAST ORIC or the library cover APCs or have open-access agreements?
8. What is the expected deadline for a first draft, and how often should I report progress?

6. One-line version

Train on older DRIAMS spectra, let the model choose which newer isolates are worth a lab test, test on a later year and on other hospitals — same data type throughout — and report how many lab tests each selection strategy saves.

Data facts checked 24 Sep 2026: DRIAMS scale and DRIAMS-A years (Weis et al., Nature Medicine 2022; Dryad record doi:10.5061/dryad.bzkh1899q). Decay figures: Wiesmann et al., J Clin Microbiol 2025. Cross-site figure: Duroux et al., GigaScience 2026.